# MINI PROJECT REPORT

Enterprise Student Record and Academic Management Platform

**Design Thinking and Product Planning**

## Unit – 1

This report is prepared based on the Unit-1 Mini Project guide provided for the Enterprise Student Record and Academic Management Platform.

# 1. Introduction

The Enterprise Student Record and Academic Management Platform is a software solution designed to manage student records and academic information within a higher education institution. The project focuses on understanding the educational domain, identifying stakeholders, analyzing user needs, defining software requirements, and preparing a product vision and roadmap before software development begins.

# 2. Business Understanding

Academic institutions manage student admission details, course registration, attendance, examination information, and academic results. The project studies these processes and identifies problems associated with manual record management. The objective is to understand the academic management process before designing the software solution.

# 3. Problem Statement

Educational institutions require an efficient way to manage student and academic information. Manual record management can make maintaining, accessing, and reporting student information difficult. The proposed platform aims to provide a structured system for student registration, record management, attendance, results, and academic reports.

# 4. Objectives

• Understand the academic management process.
• Identify stakeholders and their needs.
• Apply the five stages of Design Thinking.
• Prepare functional and non-functional requirements.
• Define the product vision and business objectives.
• Define the Minimum Viable Product (MVP).
• Prepare a phased product roadmap.
• Prepare the initial system architecture and project plan.

# 5. Design Thinking

The project applies five stages of Design Thinking: **Empathize, Define, Ideate, Prototype, and Test.** Empathize focuses on understanding users. Define identifies the main problem. Ideate generates solution ideas. Prototype creates a basic representation of the solution. Test collects feedback and identifies improvements.

# 6. IBM Design Thinking

The project applies the IBM Design Thinking principles of **Focus on User Outcomes, Restless Reinvention, and Diverse Empowered Teams**. It also documents the **Observe** → **Reflect** → **Make** cycle to support continuous understanding, reflection, and creation.

# 7. Stakeholder Analysis

The identified stakeholders are students, faculty members, academic administrators, examination cell staff, and IT support personnel. Each stakeholder has different responsibilities, challenges, and expectations from the platform.

## 8. User Personas

**Student:** Needs access to personal and academic information such as attendance and results.

**Faculty Member:** Needs to manage attendance and academic information and generate reports.

**Academic Administrator:** Needs to manage student records and academic information efficiently.

**Examination Cell Staff:** Needs to manage examination and result information.

**IT Support Personnel:** Needs to maintain and support the platform.

## 9. Requirement Gathering

**Functional Requirements:** student registration, attendance management, result generation, and report generation.

**Non-Functional Requirements:** security, usability, reliability, and scalability.

## 10. Product Vision

**Vision Statement:** To provide a centralized platform for managing student records and academic information.

**Mission Statement:** To support efficient academic record management through organized student, attendance, result, and report information.

**Success Criteria:** The platform should address identified user needs and provide the planned academic management functions.

## 11. Minimum Viable Product (MVP)

The first release of the platform includes **student registration, record management, attendance, and academic report generation**. These features form the initial usable version of the product.

## 12. Product Roadmap

The product roadmap can be organized into phases aligned with course progression: business understanding and requirements; Design Thinking and user analysis; system design and prototype; MVP development; testing and feedback; and subsequent improvements.

## 13. Initial System Architecture

A basic architecture consists of users interacting with a user interface, which connects to the application/business logic and then to the student and academic database. The system provides academic information and reports to authorized users.

## 14. Project Planning

Project planning includes designing the repository structure, initializing Git, preparing the project charter, and preparing the milestone plan. A suggested folder structure is: Documentation, Requirements, Design, Prototype, Source-Code, Testing, Reports, and README.md.

## 15. Expected Benefits

The planned platform can organize student records, simplify access to academic information, support attendance and result management, assist with report generation, and provide a foundation for Agile software development.

## 16. Project Deliverables

The project deliverables include the Business Requirements Document (BRD), Product Vision Document, Stakeholder Analysis Report, User Persona Report, Functional Requirements Specification (FRS), Non-Functional Requirements Specification (NFRS), Design Thinking Report, IBM Design Thinking Report, IBM Loop Documentation, MVP Definition, Product Roadmap, Initial System Architecture Diagram, Project Charter, Project Repository Initialization, Project Folder Structure, and Product Discovery Documentation.

## 17. Conclusion

The Enterprise Student Record and Academic Management Platform provides a planned approach to managing student and academic information. By applying Design Thinking, IBM Design Thinking, stakeholder analysis, persona development, requirement gathering, product vision, MVP planning, roadmap development, and project planning, the project establishes a foundation for Agile planning and future development.